To Detect Background Jobs, Launch Non-Blocking Processes, Return Prompt Immediately, Store Job Information, Monitor Execution, Test Background Tasks.

To Create Job Table, Store Process Groups, Track States, Update Status, Remove Completed Jobs, Validate Job Records.

```
[varnikakomali@VARNIKAs-MacBook-Air Skill % nano skill_16.c
[varnikakomali@VARNIKAs-MacBook-Air Skill % gcc skill_16.c -o skill_16 && ./skill_16
 === Part 1: Launching Background Tasks (Non-blocking & Job Table) ===

 Submitting: sleep 2 &
 [Job Created] [1] 66557 (Running in background)
 [Prompt] Shell prompt returned immediately.

 Submitting: sleep 1 &
 [Job Created] [2] 66558 (Running in background)
 [Prompt] Shell prompt returned immediately.

 Inspecting Job Table immediately after launch:
 --- Active Job Table ---
 JID      PID      PGID     STATE        COMMAND
 [1]      66557    66557    Running      sleep 2 &
 [2]      66558    66558    Running      sleep 1 &
 ------------------------

 === Part 2: State Tracking & Non-blocking Harvesting (waitpid WNOHANG) ===

 Main thread sleeping for 1.5 seconds to allow Job [2] to finish...

 Inspecting Job Table after 1.5 seconds:
 --- Active Job Table ---
 JID      PID      PGID     STATE        COMMAND
 [1]      66557    66557    Running      sleep 2 &
 [2]      66558    66558    Done         sleep 1 &
 ------------------------

 [Job Cleanup] Removing completed Job [2] (PID: 66558)

 Job Table after purging completed jobs:
 --- Active Job Table ---
 JID      PID      PGID     STATE        COMMAND
 [1]      66557    66557    Running      sleep 2 &
 ------------------------

 Main thread sleeping for another 1 second to allow Job [1] to finish...

 Final Job Table State:
 --- Active Job Table ---
 JID      PID      PGID     STATE        COMMAND
 [1]      66557    66557    Done         sleep 2 &
 ------------------------

 [Job Cleanup] Removing completed Job [1] (PID: 66557)

 Final Cleared Job Table:
 --- Active Job Table ---
 JID      PID      PGID     STATE        COMMAND
 ------------------------

 varnikakomali@VARNIKAs-MacBook-Air Skill %
```